Hacettepe University | Computer Engineering
CMP 720 EMBEDDED SYSTEMS DESIGN

Spring 2026

Midterm Project Peer Evaluation Form

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Instructions

For each project, briefly summarize the work and provide your evaluation **individually (not as a group)**. Your response should reflect your understanding of the problem, methodology, and evaluation plan, and include critical insights, strengths, limitations, and possible improvements. Skip your own group's evaluation and simply state that it is your group.

Students are required to evaluate only the projects presented within their own class group (Hacettepe or ASELSAN).

Guiding points (address in one coherent paragraph):

- What problem is being solved, why is it important, and what are the most important embedded system design considerations for this problem (e.g., timing, reliability, energy, safety, security, scalability, etc.)?
- What is the main idea of the proposed methodology?
- What do you think are the strengths and limitations of the approach?
- Is there anything you would improve or do differently?
- What is the most critical or challenging component?
- Is the evaluation plan sufficient? What, if anything, is missing?

Group 1

Umut Kalman, Eren Karadağ

Project Title: *Context-Aware Deadline Scheduling Under Memory Contention for Embedded Real-Time Systems*

Hacettepe group.

Group 2

Yusuf Çağrı Öğüt

Project Title: *Hybrid Cellular-LoRa Network Based Agriculture Monitoring and Control Application*

Hacettepe group.

Group 3

Yusuf Tahir Orhan

Project Title: *Implementation of DeFT: A Deadlock-Free and Fault-Tolerant Routing Algorithm*

Hacettepe group.

Group 4

Mehmet Yusuf Sezgi, Rana Elif Albayrak

Project Title: *Real-Time CAN Decoder*

Hacettepe group.

Group 5

Metin Irkıçatal, Anıl Arda

Project Title: *Edge-AI Based Smart Access Control System*

This project implements a real-time face recognition system on an STM32H7 MCU using an INT8-quantized MobileNetV1 model and FreeRTOS for deterministic scheduling. By leveraging Renode for simulation, the design successfully integrates image preprocessing and inference within a strict 1 MB RAM limit. Critically, while the system effectively eliminates cloud dependency, the primary challenge remains balancing the computational overhead of security features with the required sub-500ms latency for access control.

Group 6

Oğulcan Uğuroğlu, Barış Büyükyılmaz

Project Title: *Contention-Aware Duplication Scheduling for NoC Systems*

Hacettepe group.

Group 7

Ahmet Yiğitalp Demirci

Project Title: *Real-Time Energy-Constrained UAV Optimization*

This study proposes a predictive energy-time optimization algorithm for UAVs, replacing traditional static RTH thresholds with a hardware-grounded PWM-to-power model. By decoupling path planning from energy management, the system provides proactive adjustments against wind disturbances. However, the system's reliability is highly dependent on the precision of the dynamic power mapping, as inaccuracies could compromise safety-critical return missions.

Group 8

Sami Enes Yılmaz, Kadir Emre Avcı

Project Title: *SmartAquaria System*

This project utilizes a strictly layered, platform-agnostic architecture to manage aquatic life support through an event-driven Finite State Machine (FSM). The design emphasizes portability and testability by using an OSAL layer to decouple domain logic from hardware calls. While this architecture prevents "spaghetti code," the overhead of multiple abstraction layers must be carefully managed to ensure the system meets its one-second real-time sampling deadlines without data races.

Group 9

Mehmethan Ayrım

Project Title: *Energy-Aware Mixed-Criticality Scheduling*

This project explores the DFU (Dynamic Frequency Updating) algorithm for mixed-criticality systems, using MATLAB to simulate energy savings through DVFS. By reclaiming slack time from HI-task completions and sporadic arrivals, the approach aims to enhance sustainability in avionics and automotive applications. From a critical standpoint, the transition from simulation to real-world hardware must account for context-switching latencies and frequency transition times that are often idealized in MATLAB environments.

Group 10

İrem Gül Subaşı

Project Title: *Fault Injection and Resilience Profiling of Lightweight Transformers*

This research evaluates the fault resilience of quantized Transformers on FPGAs by profiling Single Event Upsets (SEUs) and establishing an architectural "Vulnerability Ranking." This ranking allows for targeted hardware hardening, such as selective Triple Modular Redundancy (TMR), to maximize resilience within limited logic resources. However, the study's focus is currently specific to lightweight models, and the generalizability of these vulnerability profiles to larger transformer architectures remains an area for further investigation.

Group 11

Süleyman Ege Karakaya

Project Title: *Adaptive Data-Movement Strategy for Crypto Accelerators*

This study introduces an adaptive data movement strategy for AES and ECDSA workloads on STM32H7-class MCUs, dynamically selecting between CPU-driven and DMA-based transfers. By considering payload size and memory contention, the project aims to optimize throughput and energy efficiency. Critically, the success of the runtime selector depends on its ability to remain lightweight; if the decision logic is too complex, it could negate the performance benefits, particularly for smaller cryptographic tasks.

Group 12

Gökçer Erol

Project Title: *Acoustic Localization System for Search and Rescue*

Hacettepe group.

Group 13

Emre Doğan, Yavuz Selim Kızıltaş

Project Title: *Optimization of Edge AI Pipeline on TDA4VM*

Focusing on the TDA4VM SoC, this project optimizes a heterogeneous Edge AI inference pipeline by distributing tasks across ARM, C7x DSP, and MMA cores. Using INT8 post-training quantization and TIDL-based scaling, it achieves high-performance object detection with minimal accuracy loss. A critical engineering challenge for this implementation is managing shared memory (L3 RAM) and bus bandwidth to prevent execution bottlenecks in the multi-core pipeline.

Group 14

Mehmet Yusuf Bilge, Sezil Ağırbaş

Project Title: *Closed-Loop Vestibular Stimulation System*

Hacettepe group.

Group 15

Ayberk Aygün, Berkay Taşkın

Project Title: *Quadrotor Simulation on NVIDIA Jetson*

Hacettepe group.

Group 16

Yasin Arslan, Süleyman Sertaç Üst

Project Title: *Real-Time Audio Pipeline Analysis on Heterogeneous SoCs*

Our group - which aims to develop a deterministic audio processing pipeline on a virtualized Zynq platform using Renode for simulation.

Group 17

Oğuz Kağan Yağlıoğlu, Emre Esef Kılıçdoğan

Project Title: *Predictive Maintenance of BLDC Motors via Multi-Sensor Fusion*

Hacettepe group.

Group 18

Hüseyin Eren Doğan

Project Title: *EdgeRAG: Offline RAG on Embedded Systems*

EdgeRAG implements an offline Retrieval-Augmented Generation pipeline for resource-constrained ARM architectures using 4-bit quantized Small Language Models like Phi-3 Mini. The system is designed to run within a strict 2 GB RAM limit in air-gapped environments, ensuring data privacy and security. The primary technical hurdle involves optimizing the memory footprint of the vector database and LLM inference to avoid Out-of-Memory (OOM) errors while maintaining acceptable response times.

Group 19

Ali Bölücü

Project Title: *Spectral-Seeded NSGA-II for Many-Core Systems*

Hacettepe group.

Group 20

İlter Doğaç Dönmez

Project Title: *Autonomous Obstacle Avoidance Vehicle using LiDAR*

Hacettepe group.